



Quantitative Portfolio Management

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What will you be able to do after this session?

- Understand risk and returns terminology
- Conceptualize the portfolio building process
- Build a portfolio with multiple stocks
- Compute portfolio risk and returns
- Profitability analysis of a portfolio and strategy

What will you learn in this session?

- Computing Returns and Risk Metrics
 - Single period return, holding period returns, annualized returns, covariance, standard deviation of returns
- Multi Stock Portfolio Building
 - Portfolio building process, modern portfolio theory, diversification effect, weights optimization
- Profitability Analysis
 - Sharpe ratio, sortino ratio, jensen's alpha, calmar ratio, information ratio, treynor ratio
- Python implementation of above-mentioned concepts



Returns

Single Period Return

- A return obtained investment in a single period; it can be a day, month, year or anything depending on what we consider a period.
- It can be calculated as follows:

$$\text{Single Period Return} = R_1 = \frac{\text{Sell price} - \text{Buy price}}{\text{Buy price}}$$

Holding Period Return

- Holding period return is the total return received from holding an asset or portfolio of assets over a period of time, generally expressed as a percentage.
- It can be calculated as follows:

$$HPR = (1 + R_1) \times (1 + R_2) \times (1 + R_3) \times \cdots \times (1 + R_n)$$

- The 4 period holding return is calculated as shown below:

$$HPR = (1 + R_1) \times (1 + R_2) \times (1 + R_3) \times (1 + R_4)$$

Other type of returns

- Risk Free Return
 - It is the return obtained on investment with zero risks. For example, it can be the rate of returns received on t-bills, certificate of deposits, fixed deposits, short term govt bonds, and so on.
- Benchmark Return
 - It is the return yielded from the benchmark index, portfolio or asset. The benchmark can be different for different assets and markets. For example, it can be a broad-based index, short term bond index, market portfolio, and so on.
- Excess Return
 - It is the return the investment has obtained above the benchmark return. It gives us an idea of how a portfolio or an asset has performed compared to a benchmark.

Annualized Returns

- It is the geometric average amount of money earned by an investment each year over a given time period. The annualized return formula is calculated as a geometric average to show what an investor would earn over a period of time if the annual return was compounded.
- It is also known as the Compounded Annual Growth Rate (CAGR).
- It is calculated as shown below:

$$\text{Annualized Returns} = (1 + \text{Cumulative Return})^{\frac{365}{N}} - 1$$

- Where:
 - **365** is the number of days in a year
 - **N** is the number of trading days for a portfolio or strategy

Portfolio Returns

- It is returns generated by holding a portfolio of the assets.
- It is the sum weighted returns of each instrument in the portfolio. For example, for a two stock portfolio, the returns are calculated as shown below:

$$\text{Portfolio Returns} = (w_A * R_A) + (w_B * R_B)$$

- Where:
 - w_A = Capital invested in stock A
 - R_A = Returns of stock A
 - w_B = Capital invested in stock B
 - R_B = Returns of stock B

Risk Measures

For Single Asset

- Variance
 - It shows the variability in the underlying variable (in our case, it is stock returns). It is calculated as shown below:

$$\text{Variance} = S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

- Standard Deviation
 - It is the square root of the variance and used as the proxy of the stock volatility.

$$\text{Standard Deviation} = S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

For Single Asset

- Annualized Volatility
 - Usually, volatility is calculated for whatever frequency of data we have. For example, using the daily bars or intraday data.
 - For most risk-adjusted analysis, we tend to use volatility on annual terms.
 - This requires us to convert the volatility to annualized volatility.
 - We can convert daily volatility to annualized volatility as shown below.

$$\text{Annualized Volatility} = S \times \sqrt{252}$$

Quantifying Relationship b/w Assets

- Covariance
 - It shows how two variables move together and its value is unbounded.

$$\text{Covariance} = \text{Cov}(x, y) = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{n - 1}$$

- Correlation
 - Like covariance, it also quantifies how two variables move together. We divide the covariance term by the product of the standard deviation of each variable to bound its output value. Hence, it ranges from -1 to 1.

$$\text{Correlation} = r = \frac{\text{Cov}(x, y)}{S_x, S_y}$$

Calculate Portfolio Risk

- We can calculate portfolio risk or volatility using the risk profiles of each of the portfolio security and their covariances.
- For a two-stock portfolio, the volatility can be calculated using the formula as shown below:

$$\text{Portfolio Variance} = w_A^2 \times \sigma^2(R_A) + w_B^2 \times \sigma^2(R_B) + 2 \times w_A \times w_B \times \text{cov}(R_A, R_B)$$

- Where:
 - w denotes weights in respective stocks
 - R depicts returns of the respective stocks
 - cov shows covariance between stocks
- For portfolios with more than two assets, we can use covariance matrix method to calculate the portfolio variance.

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- Benchmark: With what asset the portfolio should be compared to? (Which asset should the portfolio be compared)

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- Benchmark: With what asset the portfolio should be compared to?
- Any other?

Python Implementation

Python Demo

- Build a portfolio with two stocks
- Calculate portfolio returns
- Understand portfolio risk profile
- Optimize portfolio weights
 - Mean Variance Optimization
 - Monte Carlo simulations
- Understand how diversification helps

Profitability Analysis

Sharpe Ratio

- The Sharpe Ratio tells us whether the returns on a portfolio are due to good investment decisions or the result of the excessive risk taken. Higher the Sharpe ratio, the better it is.
- It gives us an idea on how the portfolio is paying us for the risk taken on the investment. It is also known as the reward to volatility ratio and is calculated as shown below:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

- Where:
 - R_p = Portfolio returns
 - R_f = Risk-free returns
 - σ_p = Portfolio standard deviation

Sortino Ratio

- Sortino ratio replaces the denominator of the Sharpe ratio with the standard deviation of the negative returns instead of the standard deviation of the total returns.
- It is a good measure in comparing the highly volatile portfolios whereas the Sharpe ratio is better at analyzing portfolios with low volatility.

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 - σ_d = Standard deviation of the negative returns

Treynor Ratio

- It is a performance measure that adjusts for systematic risk. It replaces the denominator of the Sharpe ratio with portfolio beta, which is a measure of the systematic risk.
- It is widely used to rank the identical portfolios and calculated as follows:

$$\textit{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

- Where:
 - R_p = Portfolio returns
 - R_f = Risk-free returns
 - β_p = Beta of the portfolio
- For negative values of beta, the ratio does not give meaningful values.

Calmar Ratio

- It measures the performance of an investment fund compared to its risk. It's a function of the portfolio's annualized rate of return versus its maximum drawdown.
- It is, commonly, calculated using the past 36 months data and calculated as follows:

$$\text{Calmar Ratio} = \frac{\text{Annualized}(R_p)}{\text{MaxDD}_p}$$

- Where:
 - **Annualized(R_p)** is the annualized portfolio rate of return
 - **MaxDD $_p$** is the maximum drawdown of the portfolio

Information Ratio

- It is defined as the excess returns of the portfolio relative to its tracking error. It is also known as the *appraisal ratio*.
- Often used to gauge a portfolio manager's ability to generate excess returns relative to a benchmark. It measures how close the returns of the portfolio and the benchmark are over time.

$$\text{Information Ratio} = \frac{R_p - R_b}{\sigma(R_p - R_b)}$$

- Where:
 - R_p is the portfolio return
 - R_b is the benchmark return
 - $\sigma(R_p - R_b)$ is the tracking error / Standard deviation of excess returns

The End!